

University of Cape Town
Department of Mathematics and Applied Mathematics

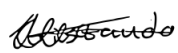
Maths Pure Honours
Galois Theory
Total marks: 80

- The deadline for submission is Thursday October 30th.
- Late submissions will not be considered.
- Email your assignment to me.
- This assignment has 3 pages of questions excluding this one.

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Question:	1	2	3	Total
Marks:	15	10	55	80
Score:				

1. Let $f(x) = x^2 + 2 \in \mathbb{Z}_5[x]$.

- (a) 4 marks Prove that f is irreducible over $\mathbb{Z}_5$.
- (b) 1 mark How many elements does $K = \mathbb{Z}_5[x]/\langle f \rangle$ have?
- (c) 10 marks Let $t = x + \langle f \rangle \in K$. Show that $1 + t$ is a generator of K^* .

My Solution:

- (a) The degree of f is 2, so if f is reducible over $\mathbb{Z}_5$, it must factorise into linear factors. But

$$\begin{aligned}f(0) &= 0^2 + 2 = 2 \\f(1) &= 1^2 + 2 = 3 \\f(2) &= 2^2 + 2 = 6 = 1 \\f(3) &= 3^2 + 2 = 11 = 1 \\f(4) &= 4^2 + 2 = 18 = 3\end{aligned}$$

and so f has no roots in $\mathbb{Z}_5$ and thus cannot be reduced into linear factors and is thus irreducible.

- (b) By the Description of Polynomial Factor rings, since f is a monic polynomial in $\mathbb{Z}_5[x]$ of degree 2,

$$\mathbb{Z}_5[x]/\langle f \rangle = \{a_0 + a_1t : a_0, a_1 \in \mathbb{Z}_5, f(t) = 0\}$$

Then $f(t) = 0 \iff t^2 + 2 = 0 \iff t^2 = -2 = 3$ and there are $5 \times 5 = 25$ elements.

- (c) K^* is the multiplicative group of non-zero elements of K and thus has 24 elements. If $0_K \notin \langle 1 + t \rangle$, then $\langle 1 + t \rangle$ is a subgroup of K^* . Since f is irreducible, we know that K is a field. This means it contains no zero divisors, so since $1 + t \neq 0$, $(1 + t)^k \neq 0$ for all $k \in \mathbb{Z}$ and so $0_K \notin \langle 1 + t \rangle$. By Lagrange's Theorem, $o(1 + t) | 24$. So $o(1 + t) \in \{1, 2, 3, 4, 6, 8, 12, 24\}$. Checking powers:

$$\begin{aligned}(1 + t)^1 &= 1 + t \neq 1 \\(1 + t)^2 &= 1 + 2t + t^2 = 4 + 2t \neq 1 \\(1 + t)^3 &= (4 + 2t)(1 + t) = 4 + 6t + 2t^2 = 10 + 6t = t \neq 1 \\(1 + t)^4 &= (4 + 2t)^2 = 16 + 16t + 4t^2 = 28 + 16t = 3 + t \neq 1 \\(1 + t)^6 &= (t)^2 = 3 \neq 1 \\(1 + t)^8 &= (3 + t)^2 = 9 + 6t + t^2 = 2 + t \neq 1 \\(1 + t)^{12} &= (3)^2 = 9 = 4 \neq 1 \\(1 + t)^{24} &= (4)^2 = 16 = 1\end{aligned}$$

So $o(1 + t) = 24$ and $\langle 1 + t \rangle = K^*$.

2. 10 marks Let F be a field of characteristic 0 and $E : F$ a normal extension. Suppose that the Galois group $\text{Gal}(E : F)$ is a cyclic group of order 30.

Prove that for every positive integer d dividing 30, there exists a unique intermediate field K_d of the extension $E : F$ satisfying that $[K_d : F] = d$.

My Solution:

First, since F has characteristic 0 and $E : F$ is normal, $E : F$ is a finite Galois extension. Let positive integer $d \mid 30$. By The Fundamental Theorem of Finite Cyclic Groups, since $\text{Gal}(E : F)$ is a cyclic group of order 30, there is an element of order $30/d$. Let σ be that element. Then $|\langle \sigma \rangle| = 30/d$. Let $K_d^* = \langle \sigma \rangle$. By The Fundamental Theorem of Galois Theory, there is some intermediate field K_d of $E : F$ corresponding to K_d^* . Moreover, we have that

$$[K_d : F] = |\text{Gal}(E : F) : K_d^*| = \frac{|\text{Gal}(E : F)|}{|K_d^*|} = \frac{30}{30/d} = d$$

All we have to show is that this intermediate field is unique. Let K be an intermediate field of the extension $E : F$ such that $[K : F] = d$. Again, by the Fundamental Theorem of Galois Theory, $|\text{Gal}(E : F) : K^*| = d$ and so $|K^*| = 30/d$. But $\text{Gal}(E : F)$ is a finite cyclic group, the Fundamental Theorem of Finite Cyclic Groups ensures that there is only one subgroup with order $30/d$. That is, $K^* = K_d^*$. The map $K \mapsto K^*$ is bijective (by the Fundamental Theorem of Galois Theory) so $K = K_d$.

3. Let $E : \mathbb{Q}$ be a field extension, where $E := \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$.

- (a) 5 marks Find the degree $[E : \mathbb{Q}]$ of the extension $E : \mathbb{Q}$.
- (b) 5 marks Give a basis of E as a vector space over $\mathbb{Q}$.
- (c) 10 marks List the elements of the Galois group $\text{Gal}(E : \mathbb{Q})$. To which (well known) group is $\text{Gal}(E : \mathbb{Q})$ isomorphic?
- (d) 10 marks For the intermediate fields K_i of the extension $E : \mathbb{Q}$ given below, determine the subgroup K_i^* of $\text{Gal}(E : \mathbb{Q})$.

$$K_1 = \mathbb{Q}(\sqrt{10}), \quad K_2 = \mathbb{Q}(\sqrt{2} + \sqrt{5}).$$

- (e) 10 marks How many intermediate fields K of the extension $E : \mathbb{Q}$ are such that $[K : \mathbb{Q}] = 4$?
- (f) 15 marks Show that if $f(x) \in \mathbb{Q}[x]$ is an irreducible (over $\mathbb{Q}$) monic polynomial of degree 3, then $f(x)$ has no roots in E .

My Solution:

- (a) Consider the following extensions:

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq E$$

By the degrees law,

$$[E : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbb{Q}(\sqrt{2}, \sqrt{3})] \cdot [\mathbb{Q}(\sqrt{2}, \sqrt{3}), \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}), \mathbb{Q}]$$

Clearly $\text{irr}(\sqrt{2}, \mathbb{Q}) = x^2 - 2$ (irreducible by Eisenstein's Criterion). This means $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and a basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}\}$. Now $\text{irr}(\sqrt{3}, \mathbb{Q}) = x^2 - 3$ (irreducible by Eisenstein's Criterion). But then if $x^2 - 3$ is reducible in $\mathbb{Q}(\sqrt{2})$, then $\deg(\sqrt{3}, \mathbb{Q}(\sqrt{2})) = 1$ and this implies $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$. So, if we can show $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ then $\text{irr}(\sqrt{3}, \mathbb{Q}(\sqrt{2})) = x^2 - 3$. If it were, then

$$\sqrt{3} = q_0 + q_1\sqrt{2}$$

Using that $\{1, \sqrt{2}\}$ is a basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. But $\sqrt{3}$ is not a rational multiple of $\sqrt{2}$ and is not rational, so $q_0, q_1 \neq 0$. This implies

$$3 = q_0^2 + 2q_1^2 + 2q_0q_1\sqrt{2}$$

But this implies $\sqrt{2}$ is rational, a contradiction. Thus $[\mathbb{Q}(\sqrt{2}, \sqrt{3}), \mathbb{Q}(\sqrt{2})] = 2$ and a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{2})$ is $\{1, \sqrt{3}\}$.

Finally, $\text{irr}(\sqrt{5}, \mathbb{Q}) = x^2 - 5$ (again, irreducible by Eisenstein's Criterion), so, by a similar argument as above, if we can show that $\sqrt{5} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3})$, then

$[\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 2$. We know, by the above results, a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$. Assume $\sqrt{5} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then

$$\sqrt{5} = a_0 + a_1\sqrt{2} + a_2\sqrt{3} + a_3\sqrt{6}$$

But this implies

$$5 = a_0^2 + 2a_1^2 + 3a_2^2 + 6a_3^2 + 2(a_0a_1\sqrt{2} + a_0a_2\sqrt{3} + a_0a_3\sqrt{6} + a_1a_2\sqrt{6} + a_1a_3\sqrt{12} + a_2a_3\sqrt{18})$$

And this implies

$$(5 - a_0^2 - 2a_1^2 - 3a_2^2 - 6a_3^2) + (2a_0a_1 + 3a_2a_3)\sqrt{2} + (2a_0a_2 + 2a_1a_3)\sqrt{3} + (2a_0a_3 + 2a_1a_2)\sqrt{6} = 0$$

But the basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is linearly independent, so each coefficient must be 0. Solving this system of equations, we need that

$$a_0 a_2 = -a_1 a_3 \quad (1)$$

$$a_0 a_3 = -a_1 a_2 \quad (2)$$

$$2a_0 a_1 = -3a_2 a_3 \quad (3)$$

$$5 = a_0^2 + 2a_1^2 + 3a_2^2 + 6a_3^2 \quad (4)$$

Multiply (1) and (2). Then

$$a_0^2 a_2 a_3 = a_1^2 a_2 a_3 \iff a_2 a_3 (a_0^2 - a_1^2) = a_2 a_3 (a_0 - a_1)(a_0 + a_1) = 0$$

From here we consider cases:

- $a_2 = 0$. Then $a_1 a_3 = a_0 a_3 = a_0 a_1 = 0$. This implies at least 2 of a_0, a_1, a_3 is zero. If all are zero, equation (4) is violated. If one is nonzero, then, by (4), it implies that element is irrational, another contradiction. So this case is impossible.
- $a_0 = a_1$. Again, by a similar argument to above, $a_1 \neq 0$. So, by (1), $a_2 = -a_3$. By (3), $2a_1^2 = 3a_2^2$. But then, by (4), $5 = a_1^2 + 2a_1^2 + 2a_1^2 + 4a_1^2$. That is, $5 = 9a_1^2$, implying a_1 is irrational, a contradiction.
- $a_0 = -a_1$. Again, by a similar argument to above, $a_1 \neq 0$. So, by (1), $a_2 = a_3$. By (3), $2a_1^2 = 3a_2^2$. Then simply proceed as the above case to get a contradiction.

So $\sqrt{5} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Finally, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 2$ (and a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is $\{1, \sqrt{5}\}$). Finally,

$$[E : \mathbb{Q}] = 2^3 = 8$$

- (b) In (a), we have found a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ (being $\{1, \sqrt{5}\}$) and a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ (being $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$). Combining these, using the proof of the Degree's Law, a basis of E over $\mathbb{Q}$ is

$$\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}, \sqrt{5}, \sqrt{10}, \sqrt{15}, \sqrt{30}\}$$

- (c) We have that

$$\text{irr}(\sqrt{2}, \mathbb{Q}) = x^2 - 2$$

$$\text{irr}(\sqrt{3}, \mathbb{Q}) = x^2 - 3$$

$$\text{irr}(\sqrt{5}, \mathbb{Q}) = x^2 - 5$$

(All irreducible by Eisenstein's Criterion). For any $\sigma \in \text{Gal}(E : \mathbb{Q})$, we must have that $\sigma(\sqrt{2})$ is a root of $x^2 - 2$. The roots of $x^2 - 2$ are $\pm\sqrt{2}$. Similarly, $\sigma(\sqrt{3}) = \pm\sqrt{3}$ and $\sigma(\sqrt{5}) = \pm\sqrt{5}$, and σ is uniquely determined by $\sigma(\sqrt{2}), \sigma(\sqrt{3}), \sigma(\sqrt{5})$. So $\text{Gal}(E : \mathbb{Q})$ has 8 elements. Define the automorphisms

$$\sqrt{2} \xrightarrow{\sigma_1} -\sqrt{2}, \quad \sqrt{2} \xrightarrow{\sigma_2} \sqrt{2}, \quad \sqrt{2} \xrightarrow{\sigma_3} \sqrt{2}$$

$$\sqrt{3} \xrightarrow{\sigma_1} \sqrt{3}, \quad \sqrt{3} \xrightarrow{\sigma_2} -\sqrt{3}, \quad \sqrt{3} \xrightarrow{\sigma_3} \sqrt{3}$$

$$\sqrt{5} \xrightarrow{\sigma_1} \sqrt{5}, \quad \sqrt{5} \xrightarrow{\sigma_2} \sqrt{5}, \quad \sqrt{5} \xrightarrow{\sigma_3} -\sqrt{5}$$

And $\sigma_1, \sigma_2, \sigma_3$ all fix $\mathbb{Q}$. Then

$$\text{Gal}(E : \mathbb{Q}) = \{\text{Id}, \sigma_1, \sigma_2, \sigma_3, \sigma_1\sigma_2, \sigma_1\sigma_3, \sigma_2\sigma_3, \sigma_1\sigma_2\sigma_3\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

- (d) We know that $K_i^* = \text{Gal}(E : K_i) = \{\sigma \in \text{aut}(E) : \sigma(u) = u \text{ for all } u \in K_i\}$. For $\sigma \in \text{aut}(E)$, σ fixes K_1 if $\sigma(\sqrt{10}) = \sqrt{10}$. But $\sigma(\sqrt{10}) = \sigma(\sqrt{2}\sqrt{5}) = \sigma(\sqrt{2})\sigma(\sqrt{5})$. So either $\sigma(\sqrt{2}) = -\sqrt{2}$ and $\sigma(\sqrt{5}) = -\sqrt{5}$, or σ fixes both $\sqrt{2}$ and $\sqrt{5}$. Now $\text{irr}(\sqrt{10}, \mathbb{Q}) = x^2 - 10$ (irreducible by Eisenstein's Criterion), so a basis of K_1 over $\mathbb{Q}$ is $\{1, \sqrt{10}\}$. So fixing $\sqrt{10}$ is the only condition needed and thus

$$K_1^* = \{\text{Id}, \sigma_2, \sigma_1\sigma_3, \sigma_1\sigma_2\sigma_3\}$$

For $\sigma \in K_2^*$, we must have that $\sigma(\sqrt{2} + \sqrt{5}) = \sqrt{2} + \sqrt{5}$. We know that $\sigma(\sqrt{2} + \sqrt{5}) = \sigma(\sqrt{2}) + \sigma(\sqrt{5}) = \pm\sqrt{2} \pm \sqrt{5}$. Thus $\sigma \in K_2^*$ must fix both $\sqrt{2}$ and $\sqrt{5}$. We must show this condition is enough to fix K_2 . We have that

$$\mathbb{Q}(\sqrt{2} + \sqrt{5}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{5})$$

On the other hand, $(\sqrt{2} + \sqrt{5})^{-1} = \frac{\sqrt{5} - \sqrt{2}}{3} \in \mathbb{Q}(\sqrt{2} + \sqrt{5})$. So $\sqrt{2}, \sqrt{5} \in \mathbb{Q}(\sqrt{2} + \sqrt{5})$ and thus $\mathbb{Q}(\sqrt{2} + \sqrt{5}) = \mathbb{Q}(\sqrt{2}, \sqrt{5})$. In the same way as we did in (a), a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}, \sqrt{5}, \sqrt{10}\}$. Thus fixing both $\sqrt{2}$ and $\sqrt{5}$ is enough to fix K_1 and so:

$$K_2^* = \{\text{Id}, \sigma_2\}$$

- (e) If $[K : \mathbb{Q}] = 4$, then, by the Degrees Law, $[E : \mathbb{Q}] = [E : K] \cdot [K : \mathbb{Q}]$ so $[E : K] = 2$. Now E is the splitting field of the polynomial $f(x) = (x^2 - 2)(x^2 - 3)(x^2 - 5) \in \mathbb{Q}[x]$. So $E : \mathbb{Q}$ is normal. Now $\text{char}(\mathbb{Q}) = 0$, and $E : \mathbb{Q}$ is a finite normal extension, so it is a Galois finite extension. But then for any intermediate field K , $[E : K] = |K^*|$. So in order for $[K : \mathbb{Q}] = 4$, we need $[E : K] = |K^*| = 2$. Since $\text{Id} \in K^*$, K^* contains exactly one other automorphism. Considering the 7 other automorphisms:

$$K = \mathbb{Q}(\sqrt{3}, \sqrt{5}) \text{ (fixed by } \sigma_1\text{)}$$

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{5}) \text{ (fixed by } \sigma_2\text{)}$$

$$K = \mathbb{Q}(\sqrt{2}, \sqrt{3}) \text{ (fixed by } \sigma_3\text{)}$$

$$K = \mathbb{Q}(\sqrt{6}, \sqrt{5}) \text{ (fixed by } \sigma_1\sigma_2\text{)}$$

$$K = \mathbb{Q}(\sqrt{10}, \sqrt{3}) \text{ (fixed by } \sigma_1\sigma_3\text{)}$$

$$K = \mathbb{Q}(\sqrt{15}, \sqrt{2}) \text{ (fixed by } \sigma_2\sigma_3\text{)}$$

$$K = \mathbb{Q}(\sqrt{6}, \sqrt{10}) \text{ (fixed by } \sigma_1\sigma_2\sigma_3\text{)}$$

So there are 7 such fields.

- (f) Let u be a root of f in some extension of $\mathbb{Q}$. Assume, to the contrary, that $u \in E$. Then

$$\mathbb{Q} \subseteq \mathbb{Q}(u) \subseteq E$$

But clearly $[\mathbb{Q}(u) : \mathbb{Q}] = \deg(u, \mathbb{Q}) = 3$, and an application of the degrees law gives us that

$$[E : \mathbb{Q}] = [E : \mathbb{Q}(u)][\mathbb{Q}(u) : \mathbb{Q}] \iff 8 = [E : \mathbb{Q}(u)] \cdot 3$$

But then $[E : \mathbb{Q}(u)] = \frac{8}{3}$, and this is impossible since $[E : \mathbb{Q}(u)]$ must be an integer